



I AM A RESEARCHER WORKING on LLM pre-training. I received my Master's degree in Artificial Intelligence from Tsinghua University in 2026, where I was advised by Prof. Chun Yuan. Before that, I earned my Bachelor's degree in Information Management and Information Systems from Beihang University in 2022. During my Master's, I was a research intern at Tencent IEG and then at Microsoft Research Asia. At MSRA, I was mentored by Yeyun Gong and Weizhu Chen.

My current research focuses on **scalable LLM pre-training and architectures**, with the aim of making large language models *more capable and more efficient to build*. I work on mixture-of-experts, conditional memory, and the stability and efficiency of training. Previously, I worked on **model merging and knowledge transfer**, such as data-free merging with task vectors and knowledge distillation. I also worked on **reasoning** through self-training and data synthesis.

PUBLICATIONS

* *Equal contribution*

Conference Papers

- [1] **Memory Grafting: Scaling Language Model Pre-training via Offline Conditional Memory**
Runxi Cheng, Yuchen Guan, Yongxian Wei, Qianpu Sun, Qixiu Li, Sinan Du, Feng Xiong, Chun Yuan, Yan Lu, Yeyun Gong
Conference on Neural Information Processing Systems (NeurIPS) 2026
TL;DR Stores a frozen pretrained model's offline hidden states for frequent n-grams as lookup memory, improving LLM pre-training over MoE and Engram baselines with limited overhead.
- [2] **Whoever Started the Interference Should End It: Guiding Data-Free Model Merging via Task Vectors**
Runxi Cheng*, Feng Xiong*, Yongxian Wei, Wanyun Zhu, Chun Yuan
International Conference on Machine Learning (ICML) 2025
TL;DR Shows that a linear layer's task vectors lie approximately in the subspace spanned by its inputs, and uses them to eliminate merging interference without any data — +10.9% over data-free baselines.
- [3] **Enhancing Logits Distillation with Plug&Play Kendall's τ Ranking Loss**
Yuchen Guan*, **Runxi Cheng***, Kang Liu, Chun Yuan
International Conference on Machine Learning (ICML) 2025
TL;DR A plug-and-play Kendall's τ ranking loss complements KL-based logit distillation by rebalancing gradients toward low-probability classes, consistently improving students on CIFAR-100, ImageNet and COCO.
- [4] **Spectral Characterization and Mitigation of Sequential Knowledge Editing Collapse**
Chi Zhang, Mengqi Zhang, Xiaotian Ye, **Runxi Cheng**, Zisheng Zhou, Ying Zhou, Pengjie Ren, Zhumin Chen
Annual Meeting of the Association for Computational Linguistics (ACL) 2026, Main Conference (Long Papers)
TL;DR Shows sequential edits collapse LLMs by disrupting dominant singular directions of pretrained weights, and proposes REVIVE, a plug-and-play filter that preserves that subspace across up to 20,000 edits.
- [5] **OptMerge: Unifying Multimodal LLM Capabilities and Modalities via Model Merging**
Yongxian Wei, **Runxi Cheng**, Weike Jin, Enneng Yang, Li Shen, Lu Hou, Sinan Du, Chun Yuan, Xiaochun Cao, Dacheng Tao
International Conference on Learning Representations (ICLR) 2026
TL;DR The first model-merging benchmark for multimodal LLMs, plus OptMerge, a data-free method that denoises task vectors to merge expert capabilities and modalities into one model.
- [6] **Advancing Open-Set Detection and Segmentation via Disentangled Representations**
Haokang Zhang, Yuchen Guan, **Runxi Cheng**, Yujiu Yang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026, Findings
TL;DR Disentangle-DINO splits textual and visual features into attribute- and category-specific components, with boundary-adaptive contrastive learning, to bridge caption-level and class-level annotations in open-set detection and segmentation.

[7] **HS-STaR: Hierarchical Sampling for Self-Taught Reasoners via Difficulty Estimation and Budget Reallocation**
Feng Xiong*, Hongling Xu*, Yifei Wang, **Runxi Cheng**, Yong Wang, Xiangxiang Chu
Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025, Main Conference — Oral
TL;DR Estimates problem difficulty with lightweight reward-guided pre-sampling, then reallocates the sampling budget to boundary-level problems — better math self-training at no extra sampling cost.

Preprints

[8] **Mixture of Neuron Experts**
Runxi Cheng, Yuchen Guan, Yucheng Ding, Qingguo Hu, Yongxian Wei, Chun Yuan, Yelong Shen, Weizhu Chen, Yeyun Gong
arXiv preprint, under review
TL;DR Decomposes MoE experts into neuron-granular experts and keeps only the top-k activated neurons in each, matching standard MoE with half of the activated MoE-layer parameters.

[9] **Multi-Task Model Merging via Adaptive Weight Disentanglement**
Feng Xiong*, **Runxi Cheng***, Wang Chen, Zhanqiu Zhang, Yiwon Guo, Chun Yuan, Ruifeng Xu
arXiv preprint, under review
TL;DR Subtracts a learned shared redundant vector so that task vectors become closer to orthogonal, reducing interference and improving multi-task model merging.

[10] **Learning to Pose Problems: Reasoning-Driven and Solver-Adaptive Data Synthesis**
Yongxian Wei, Yilin Zhao, Zixuan Hu, Li Shen, Xinrui Chen, **Runxi Cheng**, Sinan Du, Hao Yu, Chun Yuan, Dian Li
arXiv preprint, under review
TL;DR Trains a problem generator that reasons about problem design before writing, and uses solver feedback as a reward to synthesize problems adapted to the solver’s competence.

[11] **GSRender: Deduplicated Occupancy Prediction via Weakly Supervised 3D Gaussian Splatting**
Qianpu Sun, Changyong Shu, Sifan Zhou, **Runxi Cheng**, Yongxian Wei, Zichen Yu, Dawei Yang, Sirui Han, Chun Yuan
arXiv preprint
TL;DR Uses 3D Gaussian Splatting for weakly supervised 3D occupancy prediction, with a Ray Compensation module that borrows features from adjacent frames to remove duplicated predictions: +6.0 RayIoU over the previous state of the art.

Other Contributions

[12] **SIGMA: An AI-Empowered Training Stack on Early-Life Hardware**
SIGMA Team
Technical report, arXiv 2025
TL;DR An open-source training stack for early-life AI accelerators: 94.45% effective cluster utilization, and a 200B MoE model trained stably on 2,048 accelerators.

RESEARCH EXPERIENCE

	Microsoft Research Asia <i>Research Intern, Artificial Intelligence Reasoning Group</i>	FEB 2025 – JUN 2026 <i>Beijing</i>
	Tencent <i>Research Intern, IEG · Basic Research Group</i>	JUL 2024 – JAN 2025 <i>Shenzhen</i>

EDUCATION

	Tsinghua University <i>Master’s in Artificial Intelligence</i>	SEP 2023 – JUL 2026 <i>Beijing, China</i>
	Beihang University <i>Bachelor’s in Information Management and Information Systems</i>	SEP 2018 – JUL 2022 <i>Beijing, China</i>